

INDUSTRIAL WATER & WASTEWATER ENGINEERING SERIES

DEIONIZATION (DI / EDI) SYSTEM

Detailed Engineering & Operations Manual — Expanded Edition

DETAILED ENGINEERING & OPERATIONS MANUAL — EXPANDED EDITION

Water Chemistry Deep-Dives · Process Flow Diagrams · Design, O&M and Project Reference

Industrial Water & Wastewater Engineering Series · Reference Document for Design, O&M and Project Teams

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1. System Profiling & Overview

Deionization (DI) refers to the final polishing stage that removes trace residual ions from pretreated/RO/DM water to produce ultra-high-purity water, typically used where conductivity and silica specifications are far tighter than a standard two-bed DM plant can economically achieve on its own. DI is achieved either by disposable/regenerable Mixed Bed vessels or by continuous Electrodeionization (EDI) modules that use an electric field instead of chemical regenerants.

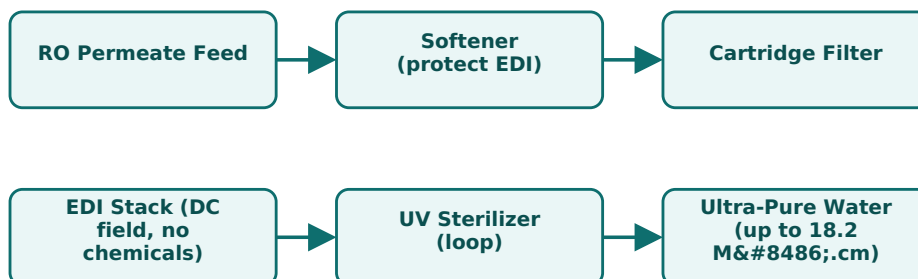


Fig. 1 — Modern RO + EDI + UV Polishing Train

1.1 Applications

- Power plant/pharmaceutical high-pressure boiler feed final polishing.
- Pharmaceutical Purified Water and Water-for-Injection (WFI) pretreatment.
- Semiconductor/electronics ultra-pure rinse water (resistivity up to 18.2 MΩ.cm).
- Laboratory-grade Type I/II water systems.
- Point-of-use portable DI cylinders for small-volume/critical applications and mobile top-ups.

1.2 DI Technology Options

Technology	Description	Typical Use
Mixed Bed DI (regenerable, in-house)	Cation + anion resin mixed in one vessel, regenerated on-site with acid/caustic	Larger continuous plants with in-house chemical handling capability
Mixed Bed DI (exchange service/portable cylinders)	Sealed resin cylinders exchanged and off-site regenerated by service provider	Small/medium demand, no on-site acid/caustic handling desired
Electrodeionization (EDI)	Continuous ion removal via DC electric field across ion-exchange membranes/resin, no chemical regeneration	Modern high-purity plants, especially pharma/power – reduces chemical handling and downtime
UV + DI Polishing Loop	UV oxidation to reduce TOC combined with DI resin polishing in recirculating loop	Pharmaceutical Purified Water distribution loops

2. Feed Water Characterization & Design Basis

Parameter	Typical Feed (post RO/DM)	Target DI Outlet
Conductivity	5-20 µS/cm	≤0.1-0.5 µS/cm (up to 18.2 MΩ.cm resistivity for UPW)
Silica	0.05-0.5 mg/L	≤0.005-0.02 mg/L
TOC (organic carbon)	<1 mg/L	<0.1-0.5 mg/L (with UV assist)
Hardness	Nil (post RO/softening)	Nil
Feed pressure	2.5-4.5 bar (typical EDI requirement)	—

3. Process Design: Mixed Bed DI & EDI

Mixed Bed DI: Cation and anion resins intimately mixed act like thousands of two-bed units in series, giving very low leakage and near-theoretical purity. Regeneration requires external separation of resins (via backwash based on density

difference), separate acid/caustic regeneration of each resin type, then re-mixing with air – a specialist operation usually done at a central regeneration facility for portable systems. Service flow rate typically 20-40 BV/hr; very sensitive to any oil/organic fouling.

Electrodeionization (EDI): RO permeate (low TDS, <25 mg/L feed recommended) passes through alternating chambers of cation and anion exchange resin/membranes under a DC electric field. Ions migrate through selective membranes toward electrodes; water splitting at the resin/membrane interface continuously regenerates the resin in-situ, eliminating chemical regeneration entirely. Requires tight upstream control of hardness (<1 mg/L as CaCO3), CO2, and chlorine to avoid scaling on concentrate side membranes and to protect resin. Typical recovery 90-95%; concentrate stream is usually recycled to RO feed to minimize water loss.

4. Pretreatment Systems

Stage	Purpose	Typical Equipment
Reverse Osmosis (upstream)	Bulk TDS reduction so DI/EDI only handles trace ionic load	Single/double-pass RO system
Softening (upstream of EDI)	Protect EDI concentrate chambers from hardness scaling	Ion-exchange softener
Micron Cartridge Filter	Protect resin/membranes from particulate fouling	1-5 micron cartridge filter
UV Sterilizer (pharma/electronics loops)	Control bioburden and TOC in recirculating loop	UV system at 254nm/185nm
Degasser (CO2 removal, if needed)	Reduce ionic load on anion resin/EDI anion chambers	Forced draft or membrane degasser

5. Operation: Regeneration vs Continuous EDI

Mixed bed DI systems operate in service-exhaustion-regeneration cycles similar to standard ion exchange, but with a critical additional step: after backwash-based resin separation, each resin type is regenerated separately and then re-mixed with compressed air to a homogeneous bed before returning to service – poor re-mixing directly degrades effluent quality. Portable exchange cylinders avoid this complexity on-site by shipping spent resin to a central regeneration facility and installing pre-regenerated replacement cylinders, trading operating simplicity for logistics dependency.

EDI, in contrast, runs continuously with no shutdown for regeneration, no acid/caustic storage or handling on site, and stable, self-limiting output quality as long as feed water specifications are maintained – making it the preferred choice in modern pharmaceutical and power-sector high-purity loops despite a higher initial capital cost.

6. High-Purity Water Chemistry Deep-Dive

6.1 Resistivity-Conductivity Relationship & Theoretical Purity Limit

Resistivity (MΩ.cm) = 1 / Conductivity (μS/cm)

Theoretical pure water limit at 25°C: 18.18 MΩ.cm (0.055 μS/cm), set by water's own self-ionization: H_2O

$\leftrightarrow \text{H}^+ + \text{OH}^-$

18.2 MΩ.cm is not an arbitrary target – it is the physical ceiling set by the ionization constant of water itself ($K_w = 1.0 \times 10^{-14}$ at 25°C). Any reading approaching this value confirms the water contains essentially no ionic contamination beyond H^+ and OH^- from water's own equilibrium; readings are temperature-compensated since K_w (and therefore baseline conductivity) increases with temperature.

6.2 EDI Water-Splitting Regeneration Mechanism

At the resin/membrane interface under the applied DC field: $\text{H}_2\text{O} \rightarrow \text{H}^+ + \text{OH}^-$

H^+ regenerates cation resin sites ($\text{R-Na} + \text{H}^+ \rightarrow \text{R-H} + \text{Na}^+$, then Na^+ migrates through cation membrane to concentrate)

OH^- regenerates anion resin sites analogously for anions

This in-situ water-splitting is what allows EDI to regenerate continuously without acid/caustic chemicals – but it only works reliably when feed hardness and CO_2 are low, because $\text{Ca}^{2+}/\text{Mg}^{2+}$ will precipitate as carbonate/hydroxide scale in the higher-pH concentrate chambers where the migrating ions accumulate, which is the chemical basis for the strict upstream softening requirement in Section 4.

6.3 TOC & UV Photo-oxidation Chemistry

185 nm UV: $\text{H}_2\text{O} + h\nu \rightarrow \cdot\text{OH} + \cdot\text{H}$ (generates hydroxyl radicals)

$\cdot\text{OH} + \text{organic carbon} \rightarrow \text{CO}_2 + \text{H}_2\text{O}$ (mineralization, oxidized further by 254nm-generated ozone/peroxide pathways)

185 nm UV light (a shorter wavelength than the 254 nm germicidal line) photolyzes water molecules to generate hydroxyl radicals that oxidize trace total organic carbon (TOC) – this is why pharma-grade Purified Water loops specify a dual-wavelength (254/185 nm) UV system rather than germicidal-only UV, since 254 nm alone controls bioburden but does not meaningfully reduce TOC.

6.4 Silica in Ultra-Pure Water

At the parts-per-billion level targeted downstream of DI/EDI, silica removal follows the same weak-acid chemistry described in the DM manual (Section 6.4 there) but is far more sensitive to trace leakage – even 0.01-0.02 mg/L residual silica is significant for semiconductor rinse water, where dissolved silica can redeposit on wafer surfaces during drying. This is why SEMI F63 electronics-grade specifications set silica limits an order of magnitude tighter than boiler-feed DM water specifications.

7. Common Problems & Engineering Solutions

Problem	Root Cause	Engineering Solution
Low resistivity/high conductivity output (Mixed Bed)	Resin exhaustion, poor re-mix after regeneration, resin fouling	Verify air-mix procedure, test resin ratio, consider replacement if fouled
EDI module low resistivity	Feed water TDS/hardness above spec, membrane scaling, electrode fouling	Check upstream RO/softener performance, verify feed conductivity and hardness
EDI current draw abnormally high	Scaling on concentrate chambers, electrode fouling	Clean-in-place (CIP) with recommended low-pH/high-pH solutions per OEM
Frequent Mixed Bed exhaustion	Higher than design feed TDS reaching DI (RO/DM upstream underperforming)	Audit upstream RO/DM performance, verify feed TDS to DI stage
TOC breakthrough in pharma loop	UV lamp aged/fouled, resin organic fouling	Replace UV lamp/clean sleeve per schedule, inspect resin condition
Air-mix incomplete after	Insufficient air scour time/pressure	Verify air-mix procedure duration and air pressure

8. Equipment Service Life

Equipment	Typical Service Life	Key Life-limiting Factors
Mixed Bed Resin	2-4 years (regenerable); cylinders exchanged per service contract	Organic/oxidative fouling, mechanical attrition from repeated separation-remix cycles
EDI Modules/Stacks	5-8 years	Membrane fouling/scaling, electrode degradation, feed water excursions
FRP/SS Vessels (Mixed Bed)	15-20 years	Corrosion, lining wear
UV Lamps	9,000-12,000 hours	UV intensity decay
DC Power Supply / Rectifier (EDI)	10-15 years	Electronic component ageing, ambient temperature/humidity exposure
Instrumentation (resistivity/TOC meters)	3-5 years	Sensor fouling, calibration drift

9. Preventive Service & Maintenance Schedule

Frequency	Activity
Daily	Monitor resistivity/conductivity trend; check EDI current and voltage; log feed water quality
Weekly	Verify UV intensity indicator; check for leaks at DI/EDI skid; inspect cylinder connections (portable systems)
Monthly	TOC testing (pharma loops); inspect EDI module for external scaling signs; review regeneration/exchange log trends
Quarterly	CIP of EDI modules if trending toward scaling; resin sample testing (in-house regenerable systems)
Annual	Full system validation (pharma/electronics); UV lamp replacement; resin/module replacement planning; recalibrate all analyzers

10. Troubleshooting Guide

Symptom	Probable Cause	Corrective Action
Resistivity dropping gradually	Resin/module nearing exhaustion, feed quality drift	Check upstream RO/softener, plan regeneration/exchange
Sudden resistivity crash	Upstream RO membrane failure, feed bypass valve open	Check RO permeate quality, verify valve line-up
EDI high reject conductivity	Poor recovery setting, concentrate recycle blocked	Verify recovery/recycle flow rates per design
High TOC in loop	Biofilm growth, UV lamp failure	Sanitize loop per SOP, replace UV lamp, review sanitization frequency
Resin cylinder exhausted faster than contract cycle	Feed water quality worse than assumed by service provider	Re-test feed water, renegotiate exchange frequency with service vendor

11. Project Execution & Engineering Workflow

Phase	Key Deliverables / Activities
Purity Requirement Study	Define target resistivity/TOC/silica/microbial specification per end-use (USP/ASTM/SEMI grade).
Basic Engineering	Technology selection (Mixed Bed vs EDI vs portable exchange), PFD, capacity sizing.
Detailed Engineering	P&ID, skid GA drawings, electrical design (EDI power supply), control philosophy, validation documentation planning (pharma).
Procurement	EDI modules/resin vessels, UV systems, instrumentation, skid fabrication.
Installation	Skid placement, piping tie-ins, electrical/control wiring, loop sanitization design (pharma).
Pre-commissioning	Flushing, hydro test, instrument calibration, dry-run of control logic.

Commissioning & Validation	Wet run, IQ/OQ/PQ documentation for pharma applications, performance verification.
Performance Test	Sustained run verifying resistivity/TOC/silica against guaranteed values at rated flow.
Handover & O&M	O&M manual, validation package, spares list, operator training.

12. Standards & Performance Guarantee

- Pharmaceutical Purified Water/WFI systems must comply with USP/EP/IP monograph limits and are subject to formal Installation/Operational/Performance Qualification (IQ/OQ/PQ).
- Electronics-grade ultra-pure water follows SEMI F63 guidelines for resistivity, TOC, particle count and microbial limits.
- Typical commercial guarantee: outlet resistivity $\geq 10\text{-}18 \text{ M}\Omega\cdot\text{cm}$ (EDI+Mixed Bed polish) or conductivity $\leq 0.1 \text{ }\mu\text{S/cm}$ at rated flow, verified over a sustained test run.
- EDI systems commonly carry a module performance warranty (voltage/current within specified band at rated recovery) separate from the overall water quality guarantee.

Technical note: figures in this manual are general design guidelines. Final technology selection and operating limits must be established from actual feed water data and end-use purity requirements.